2 Marks

1. Baye's Theorem:

If $E_1, E_2, \ldots, E_n$ are a set of mutually disjoint events with $P(E_i) \neq 0$ $(i=1,2,\ldots,n)$ then for any arbitrary event A associated with E_i is

$$P(E_i/A) = \frac{P(E_i)P(A/E_i)}{\sum_{i=1}^{n} P(E_i)P(A/E_i)}$$

2. Random Experiment:

An experiment in which the outcome cannot be fredicted but all possible outcomes is known is called as random experiment.

Eg: Throwing die is a random experiment since any one of the 6 numbers 1, 2, 3, 4, 5, 6 may appear.

3. Independent Event:

Two events are said to be independent, if the occurance of one event does not affect the probability of occurance of other.

$$P(A \cap B) = P(A) \cdot P(B)$$

4. The mean of a binomial distribution is 20 and the standard deviation is 4. Find the parameters n and p.

Soln:

Mean : $np = 20$

Variance : $npq = (SD)^2 = 4^2 = 16$

$$\frac{npq}{np} = \frac{16}{20} = 0.8$$

$$\boxed{q = 0.8}$$

$p + q = 1$.

$p = 1 - q$

$p = 1 - 0.8$

$$\boxed{p = 0.2}$$

$np = n(0.2)$

$n(0.2) = 20$

$n = 20/0.2$

$$\boxed{n = 100}$$

5. Normal Distribution :

The Normal Distribution is a continuous probability distribution characterized by a symmetric, bell-shaped curve. It is defined by two parameters, mean(μ) and standard deviation(σ).

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$$\text{for } -\infty < x < \infty$$

6. Exponential Distribution:

A continuous random variable X follows an Exponential Distribution with parameter $\lambda > 0$ if its probability density function is:

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

7. Type-I and Type-II Errors:

Type-I : Occurs when we reject the null hypothesis (H_0) when it is actually true.

Type-II : Occurs when we accept the null hypothesis (H_0) when it is actually false.

8. Cumulative Distribution Function (CDF):

The cummulative Distribution Function, $F(x)$, of a random variable X is the probability that X will take a value less than or equal to x:

$$F(x) = P(X \leq x).$$

9. Sample Space:

The sample space (denoted by S) is the set of all possible outcomes of a random experiment.

eg: In tossing a coin, $S = \{H, T\}$

where H is head and T

10. Chi-Square Distribution and its uses.

The Chi-Square (χ^2) distribution is a continuous probability distribution used for testing the significance of difference between observed (O) and Expected (E) frequency under null Hypothesis (H_0).

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

Uses:

* Goodness of fit of a distribution.
* Significance of independence of attributes.

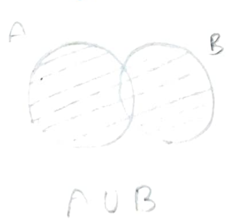
5m

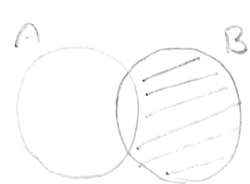
1. Addition theorem of Probability:

Let A and B be two events then,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Proof:





Here we have, $A \cup B = A \cup (\bar{A} \cap B)$

Taking probability in both sides,

$$P(A \cup B) = P(A \cup (\bar{A} \cap B))$$

Since

Since, they are disjoint,

$$P(A \cup B) = P(A) + P(\bar{A} \cap B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

We also have,

$$P(A \cap B) = P(A) + P(B) - P(A \cup B)$$

Hence proved.

2. Four cards are drawn from a well shuffled pack of cards. Find the probability that

a) All the four queens

b) There is one card from each suite

c) Two cards are diamonds and two are spades.

Soln:

Sample Space $(S) = 52C_4$

$$n(S) = \frac{52 \times 51 \times 50 \times 49}{1 \times 2 \times 3 \times 4} = 270725$$

a) Let A be the event of getting all four queens,

$$n(A) = 4C_4 = 1$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{1}{270725}$$

b) Let B be the event of getting one card from each suite

$$n(B) = 13C_1 \times 13C_1 \times 13C_1 \times 13C_1$$

$= 13 \times 13 \times 13 \times 13$

$n(B) = 28561$

$$P(B) = \frac{n(B)}{n(S)} = \frac{28561}{270725}$$

c) Let c be the event of getting two diamonds and two spades,

$n(C) = 13C_2 \times 13C_2$

$$= \frac{13 \times 12}{1 \times 2} \times \frac{13 \times 12}{1 \times 2}$$

$= 78 \times 78$

$n(C) = 6084$

$$P(C) = \frac{n(C)}{n(S)} = \frac{6084}{270725}$$

3. M.G.F of Poisson Distribution:

$$P(x) = \frac{e^{-\lambda}\lambda^x}{x!}$$

$\text{M.G.F} = E[e^{tx}]$

$$= \sum_{x=0}^{\infty} e^{tx} p(x)$$

$$= \sum_{x=0}^{\infty} e^{tx} \frac{e^{-\lambda}\lambda^x}{x!}$$

$$= e^{-\lambda} \sum_{x=0}^{\infty} \frac{e^{tx}\lambda^x}{x!}$$

$$= e^{-\lambda}\left[\frac{e^{0}\lambda^{0}}{0!}+\frac{e^{t(1)}\lambda^{1}}{1!}+\frac{e^{t(2)}\lambda^{2}}{2!}+\frac{e^{t(3)}\lambda^{3}}{3!}+\dots\right]$$

$$= e^{-\lambda}\left[1+\frac{e^{t}\lambda}{1!}+\frac{e^{2t}\lambda^{2}}{2!}+\frac{e^{3t}\lambda^{3}}{3!}+\dots\right]$$

$$= e^{-\lambda}\left[1+\frac{e^{t}\lambda}{1!}+\frac{(e^{t}\lambda)^{2}}{2!}+\frac{(e^{t}\lambda)^{3}}{3!}+\dots\right]$$

$$= e^{-\lambda}e^{e^{t}\lambda}$$

$$\text{M.G.F} = e^{-\lambda+\lambda e^{t}}$$

4. Mean and Variance of Binomial distribution.

$$P(x) = nCx\, p^{x} q^{n-x}$$

$$\text{Mean} = E(x) = \sum_{x=0}^{n} x\,p(x)$$

$$= \sum_{x=0}^{n} x\, nCx\, p^{x} q^{n-x}$$

$$= (0) + nC_1 p^{1} q^{n-1} + (2)\, nC_2\, p^{2} q^{n-2} + \dots + nC_n\, p^{n} q^{n-n}$$

$$= nC_1 pq^{n-1} + \frac{2(n)(n-1)}{1\times 2} p^{2} q^{n-2} + \dots + np^{n}q^{0}$$

$$= npq^{n-1} + n(n-1)p^{2}q^{n-2} + \dots + np^{n}$$

$$= np\left[q^{n-1} + (n-1)pq^{n-2} + \dots + p^{n-1}\right]$$

$$= np\left[q+p\right]^{n-1} \qquad \because p+q=1$$

$$= np\,(1)^{n-1}$$

$$\text{Mean} = np$$

Variance $= E(x^2) - (E(x))^2$

To find $E(x^2)$

$$E(x^2) = \sum_{x=0}^{n} x^2 p(x)$$

$$= \sum_{x=0}^{n} x^2 \, nC_x p^x q^{n-x}$$

Subtract x and add x,

$$= \sum_{x=0}^{n} (x^2 - x + x) \, nC_x p^x q^{n-x}$$

$$= \sum_{x=0}^{n} (x^2 - x) \, nC_x p^x q^{n-x} + \sum_{x=0}^{n} x \, nC_x p^x q^{n-x}$$

$$= (0) + (0) + (2^2 - 2) \, nC_2 p^2 q^{n-2} + \dots + (n^2 - n) \, nC_n p^n q^{n-n} + np$$

$$= \frac{2(n)(n-1)}{1 \times 2} p^2 q^{n-2} + \dots + (n^2 - n)(1) p^n q^0 + np$$

$$= n(n-1) p^2 q^{n-2} + \dots + (n^2 - np^n) + np$$

$$= n(n-1) p^2 q^{n-2} + \dots + (n(n-1)p^n) + np$$

$$= n(n-1) p^2 [q^{n-2} + \dots + p^{n-2}] + np$$

$$= n(n-1) p^2 [q + p]^{n-2} + np \qquad [\because p + q = 1]$$

$$E(x^2) = n(n-1)p^2 + np$$

Variance $= E(x^2) - (E(x))^2$

$$= n(n-1)p^2 + np - (np)^2$$

$$= np[(n-1)p + 1 - np]$$

$$= np[np - p + 1 - np] \qquad [\because 1 - p = q]$$

$$= np[1 - p]$$

Variance $= npq$

M.G.F of Binomial Distribution

$$P(x) = nC_x p^x q^{n-x}$$

$$M.G.F = E[e^{tx}]$$

$$= \sum_{x=0}^{n} e^{tx} p(x)$$

$$= \sum_{x=0}^{n} e^{tx} nC_x p^x q^{n-x}$$

$$= \sum_{x=0}^{n} (e^t p)^x nC_x q^{n-x}$$

$$= (1)(1) q^n + e^t p n q^{n-1} + \frac{(e^t p)^2 n(n-1)}{1 \times 2} q^{n-2} + \cdots + (e^t p)^n (1) q^0$$

$$= q^n + e^t p n q^{n-1} + \frac{n(n-1)}{2} (e^t p)^2 + \cdots + (e^t p)^n$$

$$= [q + e^t p]^n$$

$$M.G.F = [q + e^t p]^n$$

6. Properties of Normal Distribution:

* The curve is bell shaped and Symmetrical above the line $x = \mu$.

* Mean = Mediun = Mode

* As x increase, $f(x)$ will rapidly decreases as the maximum probability occurs at the point $x = \mu$

* $f(x)$ being probability will not lie on negative that means no portion of the curve lie below the X-axis.

* $\beta_1 = 0$ & $\beta_2 = 3$

* Mean Deviation = $\frac{4}{5}$ SD.

* Area Property:

$P(\mu - \sigma < X < \mu + \sigma) = 0.6826$

* X axis is an asymptote to the curve.

* Quartiles are given by $Q_1 = \mu - 0.645\sigma$ and $Q_3 = \mu + 0.645\sigma$

7. A bag contains 7 red balls, 12 white and 4 green balls. what is the probability that a) 3 balls drawn are all white, b) 3 balls drawn are one each colour

Soln:

Sample Space = $7 + 12 + 4$

$n(S) = 23C_3$

$n(S) = \frac{23 \times 22 \times 21}{1 \times 2 \times 3} = 1771$

Let A be the event of getting 3 white balls.

$n(A) = 12C_3$

$$= \frac{12 \times 11 \times 10}{1 \times 2 \times 3} = 220$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{220}{1771} = 0.124$$

Let B be the event of getting each color balls

$$n(B) = 7C_1 \times 12C_1 \times 4C_1$$

$$= 7 \times 12 \times 4$$

$$= 336$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{336}{1771} = 0.189$$

~~8. Working Procedure for testing single proportion.~~

10 M:

1. A discrete random variable has the following distribution

X	0	1	2	3	4	5	6	7	8
P(x)	a	3a	5a	7a	9a	11a	13a	15a	17a

i) Find the value of a

ii) Find $P(X > 5)$

iii) Distribution function of X

Soln :

i) $\sum_{i=0}^{n} p(x_i) = 1$

$a + 3a + 5a + 7a + 9a + 11a + 13a + 15a + 17a = 1$

$81a = 1$

$a = 1/81$

ii) $P(X > 5)$

$P(X > 5) = P(X = 6) + P(X = 7) + P(X = 8)$

$= 13a + 15a + 17a$

$= 45a$

$= 45(1/81)$

$= 0.55$

iii) The distribution function of X

X	$F(x) = P(X \le x)$
0	$P(X \le 0) = a = 0.01$
1	$P(X \le 1) = 4a = 0.04$
2	$P(X \le 2) = 9a = 0.11$
3	$P(X \le 3) = 16a = 0.19$
4	$P(X \le 4) = 25a = 0.3$
5	$P(X \le 5) = 36a = 0.44$

6	$P(X \le 6)$	$= 49a = 0.6$
7	$P(X \le 7)$	$= 64a = 0.7$
8	$P(X \le 8)$	$= 81a = 1$

2) If X is a random variable following binomial distribution with mean 2.4 and variance 1.44 find $P(X<5)$.

Soln:

Binomial Distribution $= nC_x p^x q^{n-x}$

Mean $= 2.4 = np$

Variance $= 1.44 = npq$

$npq = 1.44$

$2.4q = 1.44$

$q = \frac{1.44}{2.4}$

$\boxed{q = 0.6}$

$p = 1-q$

$= 1-0.6$

$\boxed{p = 0.4}$

$np = 2.4$

$n(0.4) = 2.4$

$n = \frac{2.4}{0.4} = 6$

$$P(x<5) = P(x=0) + P(x=1) + P(x=2) + P(x=3) + P(x=4)$$

$$P(x=0) = 6C_0 (0.4)^0 (0.6)^{6-0}$$
$$= 1(1)(0.04)$$
$$= 0.04$$

$$P(x=1) = 6C_1 (0.4)^1 (0.6)^5$$
$$= 6(0.4)(0.07)$$
$$= 0.16$$

$$P(x=2) = 6C_2 (0.4)^2 (0.6)^4$$
$$= 15(0.16)(0.12)$$
$$= 0.28$$

$$P(x=3) = 6C_3 (0.4)^3 (0.6)^3$$
$$= 20(0.06)(0.21)$$
$$= 0.25$$

$$P(x=4) = 6C_4 (0.4)^4 (0.6)^2$$
$$= 15(0.02)(0.36)$$
$$= 0.10$$

$$P(x<5) = 0.04 + 0.16 + 0.28 + 0.25 + 0.10$$

$$P(x<5) = 0.83$$

5

3. A completely randomized design with 10 plots 3 treatments gave the following results. Analyze

A	5	7	3	1
B	4	4	7	
C	3	5	1	

Soln:

H_0: There is no significant difference in the results

					Total
A	5	7	3	1	16
B	4	4	7		15
C	3	5	1		9
				T =	40

Correction factor $= \frac{T^2}{N} = \frac{40^2}{10} = \frac{1600}{10} = 160$

Sum of squares between rows $= \frac{(16)^2}{4} + \frac{(15)^2}{3} + \frac{(9)^2}{3} - 160$

$= 64 + 75 + 27 - 160$

$= 6$

Total sum of Squares $= 5^2 + 7^2 + 3^2 + 1^2 + 4^2 + 4^2 + 7^2 + 3^2 + 5^2 + 1^2 - 160$

$= 25 + 49 + 9 + 1 + 16 + 16 + 49 + 9 + 25 + 1 - 160$

$= 40$

ANOVA Table

No. of variation	Sum of Squares	D.o.f	Mean of Square	F-Test
Between rows	6	$r-1 = 2$	$\frac{6}{2} = 3$	$F = \frac{4.85}{3} = 1.61$
Residual	$40-6 = 34$	$N-1-2 = 7$	$\frac{34}{7} = 4.85$	
Total	40	9		

Calculated value = 1.61

Table value:

Table value for (2,7) at 5% = 4.74

CV (1.61) < TV (4.74)

$\therefore$ H_0 is accepted

Hence, There is no significant difference in results.

4. Mean and Variance of Poisson Distribution:

$$p(x) = \frac{e^{-\lambda}\lambda^{x}}{x!}$$

$$E(x) = \sum_{x=0}^{\infty} x\,p(x)$$

$$= \sum_{x=0}^{\infty} \frac{x e^{-\lambda}\lambda^{x}}{x!}$$

$$= e^{-\lambda} \sum_{x=0}^{\infty} \frac{x \lambda^x}{x!}$$

$$= e^{-\lambda} \sum_{x=0}^{\infty} \frac{x \lambda^x}{(x-1)!\, x}$$

$$= e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x}{(x-1)!}$$

Multiply & divide by λ

$$= e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x \lambda^{-1} \lambda}{(x-1)!}$$

$$= e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^{x-1} \lambda}{(x-1)!}$$

$$= e^{-\lambda} \lambda \sum_{x=0}^{\infty} \frac{\lambda^{x-1}}{(x-1)!}$$

$$= e^{-\lambda} \lambda \left[\frac{\lambda^{0-1}}{(0-1)!} + \frac{\lambda^{1-1}}{(1-1)!} + \frac{\lambda^{2-1}}{(2-1)!} + \dots \right]$$

$$= e^{-\lambda} \lambda \left[\frac{\lambda^0}{0!} + \frac{\lambda}{1!} + \frac{\lambda^2}{2!} + \dots \right]$$

$$= e^{-\lambda} \lambda \left[1 + \frac{\lambda}{1!} + \frac{\lambda^2}{2!} + \dots \right]$$

$$= e^{-\lambda} \lambda e^{\lambda}$$

$$= \lambda$$

Mean $= \lambda$

$$\text{Variance} = E(x^2) - [E(x)]^2$$

To find $E(x^2)$

$$E(x^2) = \sum_{x=0}^{\infty} x^2 p(x)$$

$$= \sum_{x=0}^{\infty} x^2 \frac{e^{-\lambda} \lambda^x}{x!}$$

Add and Subtract by x,

$$= \sum_{x=0}^{\infty} (x^2 - x + x) \frac{e^{-\lambda} \lambda^x}{x!}$$

$$= \sum_{x=0}^{\infty} (x^2 - x) \frac{e^{-\lambda} \lambda^x}{x!} + \sum_{x=0}^{\infty} \frac{x e^{-\lambda} \lambda^x}{x!}$$

$$= e^{-\lambda} \sum_{x=0}^{\infty} (x^2 - x) \frac{\lambda^x}{x!} + \lambda$$

$$= e^{-\lambda} \sum_{x=0}^{\infty} \frac{x(x-1)\lambda^x}{x!} + \lambda$$

$$= e^{-\lambda} \sum_{x=0}^{\infty} \frac{x(x-1)\lambda^x}{(x-2)!\,(x-1)x} + \lambda$$

$$= e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x}{(x-2)!} + \lambda$$

$$= e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x \lambda^{-2} \lambda^2}{(x-2)!} + \lambda$$

$$= e^{-\lambda} \lambda^2 \sum_{x=0}^{\infty} \frac{\lambda^{x-2}}{(x-2)!} + \lambda$$

$$= e^{-\lambda} \lambda^2 \left[\frac{\lambda^{0-2}}{(0-2)!} + \frac{\lambda^{1-2}}{(1-2)!} + \frac{\lambda^{2-2}}{(2-2)!} + \frac{\lambda^{3-2}}{(3-2)!} + \dots \right] + \lambda$$

$$= e^{-\lambda}\lambda^2\left[\frac{\lambda^0}{0!}+\frac{\lambda}{1!}+\frac{\lambda^2}{2!}+\dots\right]+\lambda$$

$$= e^{-\lambda}\lambda^2 e^{\lambda}+\lambda$$

$$E(x^2) = \lambda^2+\lambda$$

$$\text{Variance} = E(x^2)-[E(x)]^2$$

$$= \lambda^2+\lambda-[\lambda]^2$$

$$= \lambda$$

$$\text{Variance} = \lambda$$

5. A factory has two machines I and II. Machine I produces 40% of items of the output and Machine II produces 60% of items. Further 4% of items produced by Machine I are defective and 5% produced by Machine II are defective. An item is drawn at random. If the drawn item is defective, find the probability that it was produced by Machine II.

Soln:

Let E_1 be the event of selecting machine I

Let E_2 be the event of selecting machine II

A be the event of selecting defective item.

$P(E_1) = \frac{40}{100}$ $\qquad P(E_2) = \frac{60}{100}$

$P(A/E_1) = \frac{4}{100}$ $\qquad P(A/E_2) = \frac{5}{100}$

Baye's theorem,

$$P(E_i/A) = \frac{P(E_i)\,P(A/E_i)}{\sum_{i=1}^{n} P(E_i)P(A/E_i)}$$

$$P(E_2/A) = \frac{P(E_2)\,P(A/E_2)}{P(E_1).P(A/E_1) + P(E_2)P(A/E_2)}$$

$$= \frac{\frac{60}{100} \times \frac{5}{100}}{\frac{40}{100} \times \frac{4}{100} + \frac{60}{100} \times \frac{5}{100}}$$

$$= \frac{0.6 \times 0.05}{0.4 \times 0.04 + 0.6 \times 0.05}$$

$$= \frac{0.03}{0.016 + 0.03}$$

$$= \frac{0.03}{0.046}$$

$$P(E_2/A) = 0.652$$